

A Lattice Quantum Field Theory Test of the Static Escrow Postulate: 1+1D and 3+1D Falsification with Modular-Hamiltonian Partial Survival

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ABSTRACT

We test the static gravitational entropy escrow postulate $S_{nm} = |U_{grav}|/T_{Unruh}$ of Whitmer [1] using lattice quantum field theory. In 1+1 dimensions, a 295-point parameter scan of a free massless scalar at lattice sizes up to $N = 3000$ finds that the dimensionless ratio $R = S_{ent}/S_{nm}$ spans 10.56 orders of magnitude across the (L, m) grid; the mass-induced entanglement $\Delta S = S_{ent} - S_{vac}$ is uniformly negative; and the result is N -converged to four decimal places. In 3+1 dimensions, an N^3 -site scan over $N \in \{10, 12, 14, 16, 18, 20\}$ confirms the Bombelli–Srednicki area law ($S_{vac}/N^2 \rightarrow 0.0228$) and shows that R has no finite continuum limit because the bipartition entropy is dominated by the area-law vacuum term, while the mass-induced ratio $R_{\Delta} = \Delta S/S_{nm}$ is bounded by 10^{-3} across the grid. Two independent code paths agree to five decimals, ruling out implementation artifacts. Mutual information $I(M_1:M_2)$ between regions surrounding the masses decays as L^{-4} , opposite to the linear growth required by the postulate. The modular Hamiltonian content ΔK , evaluated under the Bisognano–Wichmann conjecture, exhibits the right sign and approximately recovers the BW linear asymptote $\Delta K \propto d1$ in 1+1D within a small- $d1$ window ($d1 \in [2, 6]$ at $m=1$), with prefactor approximately $1/30$ of the literal BW value. At larger $d1$ the local exponent decreases smoothly into a sublinear decay tail; the previously-published $v0.4/v0.5$ figure of " $\Delta K \propto L^{0.7}$ " is here corrected to a regime-dependent characterization, with the single-power-law exponent identified as a fitting artifact across a smooth crossover. The literal bipartition-entropy reading of the postulate is ruled out in both 1+1D and 3+1D; the modular-content interpretation is approximately correct in 1+1D within the small- $d1$ BW regime, with a calculable suppression factor as the remaining open question. A companion paper [11] reports that the 3+1D modular content does not recover the BW asymptote within the $d1$ range the lattice can resolve, with ΔK peaked at $d1 \approx 2$ and decaying as $d1^{-2}$ to $d1^{-3}$ for larger $d1$, indicating that BW recovery is dimension-dependent or, equivalently, that the rate of approach to the continuum BW asymptote is substantially slower in 3+1D than in 1+1D. The framework's horizon-limit recoveries (Bekenstein–Hawking entropy via surface gravity) are independent of these flat-space tests.

I. INTRODUCTION

The Gravitational Entropy Escrow framework [1] proposes that gravitational binding energy is identified with entropy held in escrow against the local Unruh temperature,

$$S_{esc} = |U_{grav}| / T_{Unruh} , \tag{1}$$

where U_{grav} is the Newtonian binding energy of a configuration and $T_{\text{Unruh}} = \hbar a / (2\pi c k_B)$ is the Unruh temperature corresponding to the local acceleration a . For two equal point masses m at separation L , the prediction simplifies in lattice natural units ($c = \hbar = k_B = 1$, lattice spacing = 1) to

$$S_{\text{esc}} / k_B = 2\pi m L, \quad (2)$$

with G_{eff} and the source mass cancelling identically. The framework recovers Newton's law via the Verlinde construction [2], Bekenstein–Hawking entropy at horizons via direct integration of Eq. (1) with surface gravity, and the deep-MOND acceleration scale $a_0 = c^2 \sqrt{(\Lambda/3)/(2\pi)}$ from the de Sitter horizon temperature floor.

The central claim of [1] is interpretive: that $|U_{\text{grav}}|/T_{\text{Unruh}}$ has the dimensions of and represents an entropy held in escrow against the binding. Section 7.6 of [1] explicitly identifies a derivation of Eq. (1) from quantum field theory first principles as the natural next step. A companion note [3] ruled out the natural-looking covariant local-current extension of Eq. (1) as a notational reformulation of the saturated Bekenstein bound [4]. With the dynamical extension closed off, the static identification (1) becomes the load-bearing claim that requires independent testing.

This paper reports such a test. We compute several QFT entropy quantities of a free massless scalar in the presence of two localized mass distributions and compare to the prediction (2). The tests are performed in 1+1D using Peschel's exact formula for Gaussian states [5] and extended to 3+1D using the same machinery on cubic lattices. Three independent entropy measures are tested: bipartition entanglement entropy, mutual information between regions, and modular Hamiltonian content evaluated under the Bisognano–Wichmann conjecture [6].

The headline result is that the literal identification of S_{esc} with bipartition entanglement entropy fails in both 1+1D and 3+1D. The mutual information measure confirms this independently. The modular Hamiltonian content gives a partial-survival result: right structural form, wrong magnitude. Section II describes the methodology. Section III describes the validations. Section IV reports the 1+1D result, Section V the independent confirmations from mutual information and the 3+1D continuum-limit study, Section VI the modular Hamiltonian partial-survival result. Section VII discusses the falsifiability check. Sections VIII–X interpret what the test settles, methodological notes, and conclusion.

II. METHODOLOGY

A. Lattice setup (1+1D)

We discretize a free massless scalar field on a 1+1D lattice with N sites, with Dirichlet boundary conditions at both walls (avoiding the periodic-BC zero mode). The Hamiltonian is

$$H = (1/2) \sum_i [\pi_i^2 + (\phi_{i+1} - \phi_i)^2 + m_i^2 \phi_i^2], \quad (3)$$

which writes as $H = (1/2)[\pi^T \pi + \phi^T K \phi]$ with

$$K = 2I - S - S^T + \text{diag}(m_i^2), \quad (4)$$

where S is the lattice shift operator. Two equal localized masses are placed at sites $(N-L)/2$ and $(N+L)/2$, on opposite sides of the bipartition cut at site $N/2$ and at distance $L/2$ each from the cut. Single-site delta masses are used; Section IV.D verifies that the conclusions are invariant under Gaussian smearing of the mass profile.

The parameter scan covers $N \in \{100, 300, 1000, 3000\}$, $L \in \{2, 4, 6, 10, 20, 40, 80, 160, 320, 640\}$ lattice spacings, and $m \in [10^{-3}, 10^2]$ sampled across orders of magnitude, for 295 distinct (N, L, m) points.

B. Correlators and Peschel formula

The ground state of (3) is Gaussian, with closed-form correlators

$$\langle \phi_i \phi_j \rangle = \frac{1}{2} (K^{-1/2})_{ij}, \quad \langle \pi_i \pi_j \rangle = \frac{1}{2} (K^{1/2})_{ij}. \quad (5)$$

Restricting the correlators to subsystem A (sites $[0, N/2)$) gives the reduced covariances $M_\phi = \langle \phi \phi^T \rangle|_A$ and $M_\pi = \langle \pi \pi^T \rangle|_A$. The symmetric product

$$M_{\text{sym}} = M_\phi^{1/2} M_\pi M_\phi^{1/2} \quad (6)$$

has eigenvalues $v_k^2 \geq 1/4$, and the bipartition entanglement entropy follows from Peschel's formula [5]:

$$S_{\text{ent}} = \sum_k [(v_k + 1/2) \ln(v_k + 1/2) - (v_k - 1/2) \ln(v_k - 1/2)]. \quad (7)$$

The numerical procedure is: build K from (4), diagonalize to obtain Eq. (5), restrict to subsystem A, form M_{sym} from (6), diagonalize, and apply (7). All operations use double-precision dense linear algebra.

C. Comparison metrics

Three diagnostic ratios are computed from the lattice measurements and compared to the postulate's prediction (2):

$$R_{\text{total}} = S_{\text{ent}} / S_{\text{esc}}, \quad (8)$$

$$R_\Delta = \Delta S / S_{\text{esc}}, \quad \Delta S \equiv S_{\text{ent}} - S_{\text{vac}}, \quad (9)$$

where S_{vac} is the entanglement entropy at the same N with no localized masses (a function of N and the boundary condition only). In addition, mutual information $I(M_1:M_2)$ between regions surrounding each mass and modular Hamiltonian content ΔK are introduced in Sections V and VI respectively. If the postulate (2) holds with an order-unity prefactor, the diagnostic ratios should be approximately constant of order unity across the parameter scan.

D. Lattice setup (3+1D)

The 3+1D setup uses a cubic lattice of N^3 sites with Dirichlet boundary conditions on all six faces. Two point masses are placed on the central x-axis at coordinates $(i, N/2, N/2)$ with $i = (N \pm L)/2$, with the bipartition cutting the cube along the (j,k) plane at $i = N/2$. The K-matrix has shape $N^3 \times N^3$ with Laplacian structure plus diagonal mass terms; for $N = 20$ this is 8000×8000 .

The 3+1D scan covers $N \in \{10, 12, 14, 16, 18, 20\}$, $L \in \{2, 4, 6, 8\}$, and $m \in \{0.1, 1.0, 10.0\}$. Two independent computational paths were used: a CPU implementation using NumPy with OpenBLAS multi-threading across 24 cores, and a GPU implementation using torch.linalg.eigh on an NVIDIA RTX 5090. The two paths agree on ΔS at the $N = 14$ anchor to better than 2×10^{-5} , providing independent cross-validation; see Section IX.

III. VALIDATIONS

Three validations against published results were performed before any parameter scan was treated as established.

A. Cardy–Calabrese central charge (V1)

In the absence of localized masses, the bipartition entanglement entropy of a half-line on a Dirichlet-BC lattice obeys [7]

$$S_{\text{vac}}(N/2) \approx (c/6) \ln(2N/\pi) + \text{const}_{\text{lattice}}, \quad c = 1, \quad (10)$$

for a free massless scalar (central charge $c = 1$). Successive doublings of N from 50 to 1600 give the measured values shown in Table I. The slope from successive doublings converges to $(1/6) \ln 2 \approx 0.11552$, matching the predicted central charge to better than 0.1% by $N = 1600$. The non-universal additive constant ≈ -0.030 is the expected lattice-regularization-dependent piece, stable across all four scan-grid N values to four decimal places.

TABLE I. Bipartition entanglement entropy $S_{\text{vac}}(N/2)$ vs lattice size N for a massless free scalar with Dirichlet boundary conditions, compared to the Calabrese–Cardy prediction (10) with $c = 1$. The slope $(1/6) \ln 2 \approx 0.11552$ is reproduced to better than 0.1% by $N = 1600$.

N	S_{vac} (measured)	$(1/6) \ln(2N/\pi)$	Difference
50	0.55029	0.57674	-0.02645
100	0.66413	0.69226	-0.02814
200	0.77881	0.80779	-0.02897
400	0.89392	0.92331	-0.02939
800	1.00924	1.03884	-0.02960
1600	1.12466	1.15436	-0.02971

B. Heavy-mass Dirichlet limit (V2)

In the limit $m \rightarrow \infty$, the localized-mass sites become perfect Dirichlet walls, decoupling the lattice into independent regions. The bipartition cuts the inner interval $[i_1, i_2]$ of length L at its midpoint. Calabrese–Cardy then predicts

$$S_{\text{ent}}(m \rightarrow \infty) \rightarrow (1/6) \ln(2L/\pi) + \text{const}_{\text{lattice}} . \quad (11)$$

For $L = 20$: predicted $S \rightarrow (1/6) \ln(40/\pi) - 0.030 = 0.394$; measured at $m = 100$: $S = 0.394$, agreement to four decimal places. This is an internal cross-check that the localized-mass machinery and the Peschel reduction agree at the analytically tractable limit.

C. Symbolic dimensional consistency (V3)

The escrow prediction (2) was derived symbolically using sympy and verified to be the unique dimensionally-consistent form after the cancellation of G_{eff} and the source mass. Per the methodological lessons of [3], dimensional consistency is necessary but not sufficient — the substantive checks are V1 and V2.

D. 3+1D area law (V4)

In 3+1D, the vacuum entanglement entropy across a planar cut obeys the Bombelli–Srednicki area law [8, 9]:

$$S_{\text{vac}} \approx a \cdot N^2 + \text{subleading} , \quad a \approx 0.022 . \quad (12)$$

The measured area-law coefficient S_{vac}/N^2 across $N \in \{10, 12, 14, 16, 18, 20\}$ is shown in Table II, monotonically converging from below toward an asymptote near 0.0228, consistent with published values.

TABLE II. Area-law coefficient S_{vac}/N^2 in 3+1D vs lattice size N , compared to the Bombelli–Srednicki prediction (12). The sequence is monotone-increasing toward an asymptote ≈ 0.0228 .

N	S_{vac}	S_{vac} / N^2
10	2.140	0.0214
12	3.139	0.0218

N	S _{vac}	S _{vac} / N ²
14	4.331	0.0221
16	5.708	0.0223
18	7.290	0.0225
20	9.080	0.0227

IV. RESULT: BIPARTITION ENTANGLEMENT ENTROPY IN 1+1D

A. Quantitative summary

Across the 295-point (N, L, m) parameter scan in 1+1D:

(i) The diagnostic ratio R_{total} defined in Eq. (8) ranges from 2.3×10^{-9} to 83.3 at $N = 1000$ — a span of 10.56 orders of magnitude across the (L, m) grid. The geometric mean $|R_{\text{total}}|$ is approximately 8×10^{-3} .

(ii) The mass-induced contribution $\Delta S = S_{\text{ent}} - S_{\text{vac}}$ is uniformly negative across all scanned points. Localized masses reduce entanglement rather than increasing it.

(iii) Results are converged with respect to lattice size to four decimal places for $m \geq 1$ across $N \in \{100, 300, 1000, 3000\}$; the scatter in R is not a finite-size artifact.

Table III shows R_{total} at fixed (L, m) across N. The N-independence at moderate-to-heavy m establishes that the failure is not a lattice artifact.

TABLE III. Continuum-limit behavior of R_{total} at fixed (L, m) for varying N in 1+1D. For $m \geq 1$ the ratio is N-independent to four decimal places already at $N = 100$. The result is converged.

L	m	N=100	N=300	N=1000	N=3000
4	0.01	2.6388	3.3549	4.1245	4.7797
4	0.1	0.2350	0.2678	0.2827	0.2864
4	1.0	0.00737	0.00737	0.00737	0.00737
4	10	0.000495	0.000495	0.000495	0.000495
10	1.0	0.00488	0.00488	0.00488	0.00488
40	1.0	0.00206	0.00206	0.00206	0.00206
160	1.0	—	0.000739	0.000739	0.000739

TABLE IV. R_{total} across the (L, m) grid at $N = 1000$. The values span 10.56 orders of magnitude; no rescaling can recover order-unity agreement with the postulate.

L \ m	0.001	0.01	0.1	1	10	100
2	83.3	8.25	0.563	0.00881	1.07×10^{-5}	2.27×10^{-9}
4	41.6	4.12	0.283	0.00737	0.000495	4.92×10^{-5}
10	16.7	1.65	0.115	0.00488	0.000443	4.43×10^{-5}
40	4.16	0.413	0.0301	0.00206	0.000203	2.03×10^{-5}
160	1.04	0.103	0.00844	0.000739	7.37×10^{-5}	7.37×10^{-6}
640	0.26	0.026	0.00249	0.000242	2.42×10^{-5}	2.42×10^{-6}

B. Why the literal postulate fails: the structural mismatch

The escrow prediction Eq. (2) grows linearly with both m and L without bound, with G_{eff} and the source mass exactly cancelling. The free-scalar entanglement entropy across a bipartition behaves very differently. The vacuum part S_{vac} scales as $(1/6) \ln(2N/\pi)$ — bounded for fixed N , growing only logarithmically with system size. The mass-induced part ΔS is bounded above by $O(1)$ nats in absolute value: localized masses entangle or disentangle modes only within a correlation length $\xi \sim 1/m$ of themselves, giving an $O(1)$ entropy contribution that saturates as $L \rightarrow \infty$. Heavy localized masses act as effective Dirichlet walls in the chain, decreasing the bipartition entanglement and giving $\Delta S < 0$.

The mismatch is structural, not a question of a missing prefactor: S_{esc} is an extensive quantity proportional to mL ; ΔS is intensive and bounded by topology. No constant rescaling, dimensional adjustment, or modified temperature in (1) can reconcile a bounded quantity with an unbounded one.

C. Robustness against UV artifacts

The result is invariant under the choice of how the localized mass is implemented. Single-site delta masses and Gaussian-smeared profiles with widths $\sigma \in \{1, 2, 5\}$ lattice spacings all produce the same qualitative behavior: ΔS uniformly negative, R_{total} varying across many orders of magnitude. For $m \leq 0.1$, ΔS is invariant to smearing to better than 1%. For $m \sim 1$ and larger, smearing makes ΔS more negative — a broader profile is a broader "wall," removing more entanglement. Single-site UV artifacts are not driving the result.

V. INDEPENDENT CONFIRMATIONS

Two additional tests confirm the negative result of Section IV from independent angles. The first uses an entirely different entropy measure (mutual information instead of bipartition entanglement); the second probes whether the failure persists in higher spatial dimensions.

A. Mutual information $I(M_1:M_2)$

If the prediction (2) were correctly identified with QFT entropy, a natural alternative measure is the mutual information between regions M_1 and M_2 surrounding the two localized masses,

$$I(M_1:M_2) = S(M_1) + S(M_2) - S(M_1 \cup M_2). \quad (13)$$

For free massless scalars in 1+1D, the leading contribution at large separation scales as [10]

$$I(M_1:M_2) \sim (w/L)^4, \quad (14)$$

with w the width of each region. We measured $I(M_1:M_2)$ for $w = 4$ lattice sites surrounding each mass and $L \in \{10, 20, 40, 80, 160, 320\}$. The mutual information was found to decay as L^{-4} across this range, while the postulate's prediction grows as $2\pi mL$. The diagnostic ratio $R_{\text{MI}} = I(M_1:M_2)/S_{\text{esc}}$ thus spans more than 12 orders of magnitude across the parameter scan, in the opposite direction to what the postulate requires. This is an independent confirmation that the literal postulate's linear growth is incompatible with QFT entropy measures in flat space.

B. 3+1D continuum-limit study

The 3+1D bipartition entanglement entropy was computed across $N \in \{10, 12, 14, 16, 18, 20\}$ for the (L, m) grid described in Section II.D. The vacuum area-law validation (Table II) confirmed the lattice implementation. The mass-induced results yield a structurally sharper failure than 1+1D.

Table V shows R_{total} at three representative (L, m) pairs across N . The diagnostic ratio R_{total} has no finite continuum limit: it grows as N^2 across the entire (L, m) grid, with

$$R_{\text{total}} \cdot S_{\text{esc}} = S_{\text{vac}} \quad [\text{exact, ratio } 1.0000 \text{ at every measured } N]. \quad (15)$$

The mathematical reason is transparent: the bipartition entanglement entropy in 3+1D is dominated by the area-law vacuum contribution $S_{\text{vac}} \propto N^2$, which is unrelated to the mass binding, while $S_{\text{esc}} = 2\pi m L$ is N -independent. There is no rescaling of N or of S_{esc} that produces a finite R_{total} in the continuum limit; the postulate's prediction is structurally incompatible with the dominant contribution to bipartition entanglement in 3+1D.

TABLE V. R_{total} in 3+1D at fixed (L, m) for varying N . The ratio grows as N^2 (driven by the area-law vacuum term) with no finite continuum limit. The dimensionless mass-induced ratio R_{Δ} (last column, at $N = 20$) is bounded by 10^{-3} across the grid, three orders of magnitude below the postulate's prediction $R_{\Delta} = 1$.

L	m	N=10	N=12	N=14	N=16	N=18	N=20	R_{Δ} (N=20)
2	0.1	1.71	2.50	3.45	4.55	5.81	7.22	-1×10^{-4}
2	1.0	0.171	0.249	0.345	0.455	0.581	0.722	-8×10^{-4}
4	1.0	0.0855	0.125	0.173	0.228	0.291	0.361	-2×10^{-4}

The mass-induced ratio $R_{\Delta} = \Delta S / S_{\text{esc}}$ is bounded by approximately 10^{-3} across the entire (L, m) grid — three orders of magnitude below the postulate's prediction $R_{\Delta} = 1$. ΔS is uniformly negative, with $|\Delta S| \leq 0.05$ nats, confirming that the Dirichlet-wall behavior observed in 1+1D carries over to 3+1D.

The 3+1D failure is therefore not merely a smaller-magnitude version of the 1+1D failure but a structurally distinct one. In 1+1D, R_{total} is bounded but spans 10.56 orders of magnitude because S_{ent} saturates while S_{esc} grows linearly. In 3+1D, R_{total} has no continuum limit at all because the bipartition entropy is dominated by the cut-area vacuum term, which has no relationship to the mass binding.

VI. MODULAR HAMILTONIAN CONTENT: A PARTIAL-SURVIVAL RESULT

A. Bisognano–Wichmann motivation

The Bisognano–Wichmann theorem [6] states that the modular Hamiltonian K_A of a Lorentz-invariant quantum field theory restricted to a half-space is, in vacuum,

$$K_{BW} = 2\pi \int x_{\perp} T_{00}(x) d^d x, \quad (16)$$

where $x_{\perp}$ is the perpendicular distance from the entangling surface and T_{00} is the energy density. For a localized energy m at perpendicular distance L from the cut, the linear-response prediction is

$$\Delta K_{BW} \simeq 2\pi \cdot m \cdot L, \quad (17)$$

which is exactly the form of the escrow prediction (2). The structural match is striking: both are linear in m and L with the same numerical coefficient, in the same regime (linear response to a localized energy at perpendicular distance L from a half-space cut). This raises the possibility that the postulate's correct identification is with modular Hamiltonian content rather than entanglement entropy.

B. Lattice implementation

We computed

$$\Delta K = \langle K_A^{\text{vac}} \rangle_{\text{perturbed}} - \langle K_A^{\text{vac}} \rangle_{\text{vacuum}} \quad (18)$$

across the same 1+1D lattice grid as Section IV, where K_A^{vac} is the modular Hamiltonian of the vacuum reduced to subsystem A and the perturbed state is the ground state with localized masses on. The implementation uses the Williamson normal form of the vacuum covariance matrix; details follow.

Let M_ϕ and M_π be the reduced ϕ - ϕ and π - π covariances on A. Diagonalize the symmetric product $M_\phi^{1/2} M_\pi M_\phi^{1/2} = V \text{diag}(v_k^2) V^T$, and define modular eigenvalues

$$\varepsilon_k = \ln[(v_k + 1/2)/(v_k - 1/2)] . \quad (19)$$

The modular Hamiltonian quadratic kernels in the (ϕ, π) basis are then

$$Q = M_\phi^{-1/2} V \text{diag}(\varepsilon_k v_k) V^T M_\phi^{-1/2} , \quad (20)$$

$$P = M_\phi^{1/2} V \text{diag}(\varepsilon_k/v_k) V^T M_\phi^{1/2} , \quad (21)$$

such that $K_{A_quad} = (1/2)\phi^T Q \phi + (1/2)\pi^T P \pi$ plus a constant offset fixed by $\langle K_{A_quad} \rangle_{vac} = S_{vac}$. The expectation value of K_{A_quad} in any other Gaussian state with covariances M'_ϕ, M'_π is then

$$\langle K_{A_quad} \rangle = 1/2 \text{tr}(Q M'_\phi) + 1/2 \text{tr}(P M'_\pi) . \quad (22)$$

The implementation was internally validated by three checks: (i) relative-entropy positivity, $S_{rel} = \Delta K + S_{vac} - S_{pert} \geq 0$ verified for all scanned states (this is the strict consequence of relative-entropy non-negativity and is a sanity check on the modular kernel construction independent of whether the BW conjecture holds); (ii) modal-vs-original-basis cross-check, with $\langle K_{quad} \rangle_{vac}$ computed two independent ways agreeing to six decimal places; (iii) constant-offset consistency between the modular Hamiltonian's quadratic part and the vacuum entanglement entropy.

An initial implementation produced relative-entropy violations of order 10^4 , immediately flagging an error in the Williamson decomposition; correction yielded order-unity values with positivity verified throughout. See Section IX.

C. Result: right form, wrong magnitude

The lattice measurement of ΔK confirms the BW structural form: ΔK is positive, grows with L , and is approximately linear in m for moderate values. This is the correct sign and the correct scaling direction predicted by the escrow postulate when reinterpreted as modular content. Two quantitative discrepancies emerge.

First, the L -scaling is not a single power law. The published v0.4 of this paper reported a single-power-law fit $\Delta K \propto L^{0.7 \pm 0.05}$ at $(m=1, N=1000)$; a re-examination using sliding-window fits across the same data shows that this single exponent was an artifact of fitting across a smooth crossover. The local exponent, computed in 5-point sliding windows over $d1$ (the perpendicular distance of the mass in subsystem A to the cut, which equals $L/2$ in the symmetric two-mass placement), evolves continuously from

$$\Delta K \propto d1^{1.02 \pm 0.05} \quad [d1 \in [2, 6]] \quad (23)$$

at small $d1$ — indistinguishable from the BW linear asymptote — down through intermediate values to

$$\Delta K \propto d1^{0.5 \pm 0.04} \quad [d1 \in [16, 64]] \quad (23a)$$

in the decay tail. The published "0.7" was an effective exponent obtained by fitting a single power law across both regimes; it is not a universal scaling. The robust statements supported by the data are: (i) the BW linear asymptote $\Delta K \propto d1$ is approximately recovered at small $d1$ in 1+1D, with a suppressed prefactor; (ii) the crossover into a sublinear decay tail is smooth, not abrupt; (iii) the local exponent at any given window depends on which window is fit. We retract the v0.4 framing of "sublinear scaling" as a universal description and replace it with the regime-dependent characterization above.

Second, the magnitude is too small. In the small- d_1 BW regime [$d_1 \in [2, 6]$, $m=1$, $N=1000$], the prefactor $\Delta K / (2\pi \cdot m \cdot d_1)$ is approximately 0.030–0.037, corresponding to a suppression factor of roughly 1/30 relative to the literal BW prediction. The closely-related $v_{0.4}$ figure of "approximately 55× smaller at $L=4$, $m=1$ " came from the two-mass protocol where the relevant variable is $d_1 = L/2 = 2$; under either convention the magnitude suppression is on the order of 30× to 55× depending on which point in the scan is chosen. For most of the (L, m) grid at heavier mass the apparent ratio drops further (down to 10^{-6} in the heavy-mass / large- L regime), but this is largely because heavy mass takes the system outside the linear-response regime where the BW formula was derived; ΔK saturates rather than continuing to grow as $2\pi \cdot m \cdot d_1$. The clean BW comparison is at moderate mass ($m \lesssim 1$) and moderate d_1 ($d_1 \in [2, 8]$), where the linear asymptote is recovered with a suppressed prefactor.

D. Interpretation

The modular Hamiltonian result in 1+1D is consistent with the BW asymptote at small d_1 , with a suppressed prefactor. Specifically, in the regime $d_1 \in [2, 6]$ at $m = 1$, the local exponent is $+1.02 \pm 0.05$ (indistinguishable from the BW linear scaling $\Delta K \propto d_1$) and the prefactor is approximately 0.030 of the literal BW value $2\pi \cdot m \cdot d_1$. The smooth crossover into a sublinear decay tail at $d_1 \gtrsim 8$ reflects geometric corrections to the BW half-space assumption: the lattice is not a true Rindler wedge, the boundary conditions are Dirichlet rather than free, and at d_1 comparable to subsystem-A depth the half-space approximation breaks down. The decay tail is therefore consistent with a finite-size deformation of BW rather than a fundamental failure. Determining whether the prefactor 1/30 has a calculable origin (e.g., from boundary-condition corrections, from the lattice regularization, or from a missing dimension-dependent geometric factor) and whether the BW asymptote can be cleanly extracted in the limit of a true half-chain bipartition is left for future work. The result we report here is that the BW modular Hamiltonian asymptote is qualitatively correct in 1+1D within a finite small- d_1 window; this is closer to a partial confirmation than $v_{0.4} / v_{0.5}$ framings of the same data suggested.

The dimensional dependence of this result is investigated in a companion paper [11], which extends the same Williamson-decomposition methodology to a 3+1D cubic lattice. The 3+1D result is qualitatively different from the 1+1D result reported here: the BW linear asymptote is not recovered within the d_1 range the lattice can resolve, and ΔK is peaked at $d_1 \approx 2$ with a power-law decay $\Delta K \propto d_1^{-2 \text{ to } -3}$ for $d_1 \geq 3$. At matched mass $m = 1$ (within the linear-response regime where the BW comparison is strictly meaningful), the 3+1D peak ratio $\Delta K / (2\pi \cdot m \cdot d_1) \approx 0.0026$ is approximately 14× smaller than the 1+1D peak ratio ≈ 0.037 reported in this paper, so 1+1D is closer to BW than 3+1D in both qualitative scaling and absolute prefactor. The companion paper notes that the rate of approach to the continuum BW asymptote may genuinely be slower in 3+1D (where the conformal-symmetry group of the massless free scalar is finite-dimensional) than in 1+1D (where it is the infinite-dimensional Virasoro algebra), so 3+1D BW recovery at much larger d_1 inaccessible to the present lattice cannot be strictly ruled out. We refer the reader to [11] for the full 3+1D analysis and the comparative discussion.

On the calculable origin of the 1/30 prefactor in 1+1D, one candidate explanation is worth flagging as a target for follow-up theoretical work. The Bisognano–Wichmann formula $K_A^{\text{vac}} = 2\pi \int_{x_-}^x T_{00} dV$ is an infrared / long-wavelength prediction valid in the continuum where Lorentz invariance is exact. A localized lattice mass perturbation with $m \approx 1$ in lattice units corresponds to a correlation length $\xi \sim 1/m \sim 1$ lattice spacing, which is a UV-scale perturbation by the lattice's own measure. In this regime the BW formula's continuum derivation no longer strictly applies; the effective lattice modular Hamiltonian is expected to differ from the BW formula by a regulator-dependent prefactor that depends on the ratio of correlation length to lattice spacing. The 1/30 suppression observed at small d_1 may reflect this UV/IR mismatch rather than a fundamental modification of the postulate, and could in principle be calculated from first principles by computing the lattice modular Hamiltonian directly in the continuum scaling limit. We do not pursue this calculation here but identify it as a sharp theoretical question that would either remove or reify the apparent suppression. A second candidate is that the Dirichlet boundary

conditions used for the lattice differ from the free boundary conditions assumed in the standard BW derivation; the difference may contribute a calculable boundary-correction factor.

We do not adjudicate among these readings. The result is reported as the most promising direction for future modification of the postulate, and as a sharp constraint on what such modifications must satisfy.

VII. FALSIFIABILITY CHECK

A central concern when comparing computed quantities to dimensional-analysis predictions is the possibility that any apparent agreement (or disagreement) is a tautology fixed by the choice of dimensional combinations rather than reflecting physical content. We check this directly.

The Unruh temperature T_{Unruh} varies across the (L, m) grid as $T_{\text{Unruh}} \propto m/L^2$ in lattice units, spanning approximately five orders of magnitude across the scanned range. If the test were dimensionally trivial — if any choice of T would produce $R \approx 1$ — then the construction would be uninformative regardless of the result. We verified that no choice of T produces a uniform $R \approx 1$ across the parameter scan; T_{Unruh} is not privileged in this sense. The 10.56 orders of magnitude variation in R is therefore a real measurement of the disagreement between the postulate's prediction and the actual QFT entanglement structure, not an artifact of dimensional rescaling.

This is the analog of the lesson from [3]: a construction that achieves agreement only by choosing the dimensional combination that produces the desired answer is not an empirical confirmation. The lattice test passes this falsifiability check; the Section IV result is a genuine quantitative measurement of the postulate's failure as bipartition entanglement entropy.

VIII. WHAT THIS SETTLES, AND WHAT IT DOES NOT

A. *What this settles*

The literal static escrow postulate (1), taken as a quantitative identification of S_{esc} with QFT bipartition entanglement entropy in flat space with point-mass binding, does not survive in either 1+1D or 3+1D. In 1+1D, the qualitative behavior is wrong (sign, scaling, boundedness in N), the magnitude is wrong (10.56 orders-of-magnitude scatter), and the lattice-size convergence rules out finite-size explanations. In 3+1D, the bipartition entropy is dominated by the area-law vacuum term and has no finite continuum limit when divided by the postulate's prediction; the dimensionless mass-induced ratio R_{Δ} is bounded by 10^{-3} , three orders of magnitude below the postulate's prediction. Two independent code paths agree to five decimals on the 3+1D calculation, ruling out implementation artifacts. The mutual information measure confirms the failure from a third independent direction. No constant rescaling of S_{esc} by a dimensionless factor or by a modified temperature can repair the disagreement.

B. *What this does not settle*

The horizon limit is unaffected. The framework's recovery of Bekenstein–Hawking entropy at black hole horizons [1, §5] goes through surface gravity and the Unruh temperature at the horizon directly, not through volume-bipartition entanglement. The lattice test does not address the horizon recovery. The horizon claims of [1] are independent of the present result.

Curved-spacetime regimes are also outside scope. The 3+1D test reported here uses a flat-space lattice with imported Newtonian binding-energy form. The framework's predictions for genuine curved-spacetime configurations (non-perturbative gravity, full general-relativistic treatment) are not addressed by this test and require lattice quantum gravity techniques beyond the scope of the present calculation.

The framework's empirical content is unaffected. The empirical predictions of [1] — the constant- a_0 SPARC reanalysis, the deep-MOND asymptote $a_0 = c^2\sqrt{(\Lambda/3)/(2\pi)}$, and the baryonic Tully–Fisher relation — do not depend on the literal bipartition-entanglement reading of the static postulate. They depend on

the deep-MOND asymptote and the framework's interpretive identification at horizons, neither of which is refuted by the present result.

Modular-content variants remain partially open. The modular Hamiltonian result of Section VI shows the BW linear asymptote approximately recovered at small d_1 in 1+1D, with prefactor approximately $1/30$. The companion paper [11] extends the same methodology to 3+1D and finds that the BW asymptote is not recovered within the d_1 range the lattice can resolve, with ΔK peaked at $d_1 \approx 2$ and decaying as d_1^{-2} to d_1^{-3} for $d_1 \geq 3$. At matched mass ($m = 1$) the 1+1D peak ratio is also approximately $14\times$ larger than the 3+1D peak ratio, so 1+1D is closer to BW in both qualitative scaling and absolute prefactor. Taken together, the two papers show that the modular-content interpretation of the static escrow postulate has dimension-dependent validity within the lattice sizes presently accessible: approximately correct in 1+1D within a finite small- d_1 window; not recovered in 3+1D at the d_1 range probed, with the open question of whether 3+1D BW recovery can be achieved at much larger d_1 not currently accessible. Whether the 1+1D $1/30$ prefactor has a calculable origin (boundary-condition corrections, lattice regularization with UV/IR mismatch, or a missing geometric factor in the postulate) and whether the 3+1D structure can be reconciled with a modified BW form remain open theoretical questions for future work.

C. Implication for the framework

The framework's interpretive content survives the test. Its quantitative literal interpretation — specifically, the reading in which $|U_{\text{grav}}|/T_{\text{Unruh}}$ equals the bipartition entanglement entropy of a quantum field across a surface separating two masses — does not. What survives is dimension-dependent: in 1+1D the BW modular Hamiltonian asymptote is approximately recovered within a small- d_1 window, with a calculable suppression factor of approximately $1/30$ as the remaining open question; in 3+1D the BW asymptote is not recovered. The framework now lives as a horizon-thermodynamic interpretation supplemented by a 1+1D modular-content match, with the dimensional dependence of the modular-content match being the principal open problem. Section VI's small- d_1 BW recovery in 1+1D, together with the dimensional-contrast result in [11], identifies the prefactor calculation and the dimensional-dependence question as the sharpest research directions.

IX. METHODOLOGICAL NOTES

The calculations reported here were performed via autonomous runs of a large language model on local hardware (NVIDIA RTX 5090, Intel Core Ultra 9, 256 GB DDR5) with full code execution and persistence-to-disk capabilities. Three features of this methodology are worth highlighting.

First, validations against published results, not internal consistency checks alone, are the binding constraint. The Calabrese–Cardy formula (10) and the Bombelli–Srednicki area-law coefficient (12) are external benchmarks; agreement to better than 0.1% (1+1D) and a few percent (3+1D) is a meaningful confirmation that the lattice implementations are correct. Pure symbolic or dimensional checks alone cannot establish this. The methodological lesson from the C8 episode of [3] — dimensional consistency is necessary but not sufficient — was applied here from the start.

Second, the falsifiability check (Section VII) is load-bearing. Without verifying that T_{Unruh} is privileged in producing the comparison, the result could be dismissed as a tautology fixed by dimensional analysis. The check rules this out: the test has empirical content.

Third, modular-content code requires special validation. The ΔK computation of Section VI passed three internal cross-checks: relative-entropy positivity, modal-vs-original-basis agreement to six decimals, and constant-offset consistency. An initial implementation produced relative-entropy violations of order 10^4 , an unphysical result that immediately flagged the implementation error. Correction yielded order-unity values with positivity verified throughout. Relative-entropy positivity is the right sanity check for any modular-content code regardless of whether the underlying conjecture is true.

In addition, the 3+1D continuum-limit study was performed in two independent autonomous runs that took different optimization paths: one used CPU NumPy with OpenBLAS multi-threading across 24 cores and finished in approximately 48 minutes, while the other ported the eigendecomposition to GPU using torch.linalg.eigh on the RTX 5090 and finished in under 3 minutes. The two runs agreed on ΔS to better than 2×10^{-5} and on the area-law sequence S_{vac}/N^2 to four decimals. Independent code paths on different hardware reaching the same numerical answer rules out implementation artifacts and round-off-pattern artifacts simultaneously.

X. CONCLUSION

The literal static escrow postulate of [1], interpreted as a quantitative identification of $S_{\text{esc}} = |U_{\text{grav}}/T_{\text{Unruh}}|$ with QFT bipartition entanglement entropy, fails in both 1+1D and 3+1D. In 1+1D the failure is qualitative (wrong sign, Dirichlet-wall behavior), quantitative (10.56 orders of magnitude in R across the scan), and robust (lattice-size converged, invariant under localization-scheme choice, not a dimensional artifact). In 3+1D the failure is structurally sharper: R_{total} has no finite continuum limit because the bipartition entropy is dominated by the Bombelli–Srednicki area law, while S_{esc} remains N -independent; the dimensionless mass-induced ratio R_{Δ} is bounded by 10^{-3} across the (L, m) grid. Two independent code paths agree to five decimals on the 3+1D calculation. Mutual information confirms the failure from a third independent direction (R_{MI} spans 12+ orders of magnitude).

One result is now positive within a finite regime: the modular Hamiltonian content ΔK , evaluated under the Bisognano–Wichmann conjecture, approximately recovers the BW linear asymptote $\Delta K \propto d_1$ in 1+1D within a small- d_1 window ($d_1 \in [2, 6]$ at $m=1$), with a suppressed prefactor of approximately $1/30$ of the literal BW value. The published v0.4/v0.5 framing of "sublinear scaling at exponent 0.7" is here corrected to a regime-dependent characterization: the BW asymptote is recovered at small d_1 with $\alpha \approx 1.0$, the local exponent decreases smoothly through intermediate values, and a single power-law fit averages across this crossover into the apparent 0.65–0.7 effective exponent. The 1+1D BW recovery is the closest-to-positive empirical evidence in the lattice tests, with the prefactor calculation as the sharpest open question (a candidate explanation in terms of UV/IR mismatch is discussed in Section VI.D). The companion paper [11] reports that the analogous 3+1D test does not recover the BW asymptote within the d_1 range the lattice can resolve, with a peak ratio at matched mass approximately $14\times$ smaller than the 1+1D peak ratio; whether 3+1D BW recovery can be achieved at much larger d_1 inaccessible to the present lattice remains an open question.

The published framework's interpretive content and empirical results are not refuted; what is ruled out is the literal bipartition-entanglement reading of its central postulate. The framework's recovery of Bekenstein–Hawking entropy at horizons, its deep-MOND asymptote, and its constant- a_0 empirical content remain unaffected. Future tests should be directed at the modular-content reinterpretation, the horizon limit directly, or curved-spacetime regimes that go beyond the flat-space lattice approach used here.

The framework now lives as a sharper, narrower, and dimension-aware set of claims: horizon thermodynamics is its strongest empirical anchor; flat-space bipartition entanglement is ruled out as a literal identification in both 1+1D and 3+1D; modular Hamiltonian content approximately recovers the BW linear asymptote in 1+1D within a small- d_1 window (with a $1/30$ prefactor whose origin is an open theoretical question, plausibly attributable to UV/IR mismatch between the lattice-localized mass and the continuum BW formula); modular-content recovery is dimension-dependent within the lattice sizes presently accessible, with 3+1D failing to recover BW at the d_1 range probed in the companion paper [11]. Each of these is a bounded research direction with sharp positive-or-negative outcomes available from future tests, including 2+1D as a natural intermediate dimensional case and $N \gtrsim 20$ GPU runs in 3+1D as a way to test whether BW recovery is reached at larger d_1 .

VERSION NOTES

v0.7 (May 2026) updates v0.6 with three small additions following pre-deposit review by Perplexity Deep Research. (i) Section VI.D adds a paragraph speculating on the calculable origin of the $1/30$ prefactor in 1+1D as a UV/IR mismatch between the lattice-localized mass perturbation (correlation length $\xi \sim 1/m \sim 1$ lattice spacing at $m = 1$) and the continuum BW formula (an infrared / long-wavelength prediction). The Dirichlet boundary conditions are also flagged as a candidate contributor. This paragraph identifies a sharp theoretical question (a first-principles calculation of the lattice-regularized modular Hamiltonian) that would resolve the prefactor's origin. (ii) References to the companion paper [11] in the abstract, Section VI.D, Section VIII.B, and Section X have been updated for consistency with the companion paper's v0.2 framing, which compares 1+1D and 3+1D at matched mass $m = 1$ (within the linear-response regime where the BW comparison is strictly meaningful) rather than at mismatched masses. The corrected matched-mass comparison shows the 1+1D peak ratio is approximately $14\times$ larger than the 3+1D peak ratio in absolute prefactor, in addition to showing different qualitative scaling. (iii) The description of the 3+1D companion paper's result has been nuanced from "BW asymptote not recovered at any d_1 the lattice can resolve" to "BW asymptote not recovered within the d_1 range the lattice can resolve," acknowledging that the rate of approach to the continuum BW asymptote may genuinely be slower in 3+1D (where the conformal-symmetry group is finite-dimensional) and that BW recovery at much larger d_1 inaccessible to the present lattice cannot be strictly ruled out. No numerical results are altered between v0.6 and v0.7; only the discussion sections.

v0.6 (May 2026) updated v0.5 with one substantive correction. (i) The Section VI.C reporting of the ΔK L-scaling was rewritten. The previous v0.4/v0.5 statement that ΔK shows "sublinear scaling at exponent 0.7 ± 0.05 " was the result of fitting a single power law across the seven-point L scan in the two-mass protocol. Sliding-window analysis of the same data, supplemented by a single-mass-in-A protocol that cleanly disentangles d_1 (perpendicular distance of the mass in subsystem A to the cut) from L (mass separation), shows that the function $\Delta K(d_1)$ is not a single power law: the local exponent evolves smoothly from $\alpha \approx +1.0$ in a small- d_1 window ($d_1 \in [2, 6]$ at $m=1$) down through intermediate values to $\alpha \approx +0.5$ in a decay tail at $d_1 \gtrsim 16$. The published "0.7" was an effective exponent obtained by averaging across this smooth crossover, not a universal scaling. The corrected interpretation is that the BW linear asymptote $\Delta K \propto d_1$ is approximately recovered in 1+1D at small d_1 , with a suppressed prefactor of approximately $1/30$ of the literal BW value. The crossover into a sublinear decay tail at larger d_1 reflects geometric corrections to the BW half-space assumption (Dirichlet boundary conditions, finite subsystem-A depth) rather than a fundamental failure of the modular-content interpretation. (ii) The abstract, Section VI.D interpretation paragraph, and the conclusion in Section X were rewritten for internal consistency with the corrected Section VI.C, including a cross-reference to the companion paper [11] which extends the same methodology to 3+1D. (iii) The v0.5-promised 3+1D ΔK follow-up was completed and is reported as the companion paper [11]. No numerical results are altered between v0.5 and v0.6; only the framing and analysis of the existing data.

v0.5 (May 2026) updated v0.4 with two changes addressing pre-deposit referee feedback. (i) The acknowledgment language describing LLM use was tightened: the previous wording — "Three independent reviewers (Anthropic Claude Opus 4.7, xAI Grok, Perplexity Deep Research) reviewed the lattice run output and converged on the same headline assessment" — risked being read as a claim that LLM critique substituted for human peer review. The corrected wording in the Acknowledgments characterized these uses as LLM-assisted code review and cross-comparison, with the substantive cross-validation being the agreement to better than five decimal places between two independent code paths (CPU NumPy and GPU PyTorch) on different hardware, as reported in Section IX. (ii) The L-scaling exponent for ΔK reported in Section VI.C was qualified: the v0.4 statement " $\Delta K \propto L^{0.7}$ " had been reported without uncertainty or universality caveats. v0.5 reported the exponent as 0.7 ± 0.05 from the seven-point fit. v0.6 (above) supersedes this characterization with the regime-dependent analysis.

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LLM-assisted code review and cross-comparison of the lattice run output were performed by Anthropic Claude Opus 4.7, xAI Grok, and Perplexity Deep Research; small numerical details were corrected through this cross-comparison process. These tools were used for code review and consistency checks, not as substitutes for peer review. The substantive cross-validation reported in this paper is the agreement to better than five decimal places between two independent code paths (CPU NumPy with OpenBLAS and GPU PyTorch with CUDA) implementing the 3+1D continuum-limit study on different hardware, as described in Section IX. The author is solely responsible for the conclusions and any remaining errors. This work is the third in a series following [1] and [3].

DATA AND CODE AVAILABILITY

Lattice scripts, validation code, parameter-scan results (1+1D scan: 295 points at N up to 3000; 3+1D scans: original 25-point preliminary scan and the GPU-accelerated continuum-limit extension with 45 mass points + 4 vacuum entropies at $N \in \{14, 16, 18, 20\}$), modular Hamiltonian code with three internal cross-checks, mutual information code, 3+1D lattice code with both CPU and GPU paths, continuum-limit table generator, diagnostic plots, and run logs are deposited alongside this preprint. The scripts use NumPy, SciPy, and (for the GPU path) PyTorch with CUDA. A clean rerun of the full 1+1D pipeline takes approximately 7 minutes on the reference hardware; the GPU-accelerated 3+1D continuum-limit scan completes in approximately 3 minutes.

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